

Covariant Effective Field Theory for Entropy-Driven Gravitational Modification: Derivation, Falsification, and Structural Results from a 17-Iteration Construction Programme

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Abstract. We present the complete results of a systematic 17-iteration programme to construct and test a covariant scalar–tensor effective field theory (EFT) in which gravitational dynamics are modified by an entropy-like scalar field $\mathcal{S}$ via an acceleration-dependent response function. The programme established five structural results that hold independently of the specific modification function $\mu(g)$: (i) gravitational wave speed equals c exactly (shown as a theorem for minimally coupled actions of the form $\mathcal{L} = R/16\pi G + F(g_{\mu\nu}, \mathcal{S}, \partial\mathcal{S})$ containing no metric derivatives in F); (ii) gravitational slip $\eta = \Phi/\Psi \approx 1$ with deviations suppressed by $\mathcal{O}(\Phi \cdot \mu)$ in the weak-field regime, via a structural cancellation in the radial stress; (iii) solar-system constraints are satisfied by exponential amplitude suppression far below current experimental bounds; (iv) the theory is ghost-free, tachyon-free, and hyperbolic; (v) the empirical Radial Acceleration Relation is formally identical to a Bose–Einstein occupation number, $\mu = 1/(\exp(\sqrt{g/a_0}) - 1)$. A critical negative result is also reported: the classical field equation from the action $-\beta|\partial\mathcal{S}|^2$ produces a gravitational correction that *increases* with acceleration, whereas the RAR requires a correction that *decreases*. This identifies a precise structural gap requiring either quantum treatment of the $\mathcal{S}$ -field or a fundamentally non-local coupling. We formulate the surviving framework as an EFT with $\mu(g)$ postulated as effective relation, preserving all structural properties with one measured parameter $a_0 = 1.2 \times 10^{-10} \text{ m/s}^2$.

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1 Introduction

1.1 Motivation

The Radial Acceleration Relation (RAR) observed in rotationally supported galaxies establishes a tight, universal correlation between observed gravitational acceleration g_{obs} and that predicted from baryonic matter alone g_{bar} [1, 2]. The empirical interpolation function involves a single scale $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$:

$$g_{\text{obs}} = \frac{g_{\text{bar}}}{1 - \exp(-\sqrt{g_{\text{bar}}/a_0})}. \quad (1)$$

Defining the gravitational enhancement $\mu \equiv g_{\text{obs}}/g_{\text{bar}} - 1$:

$$\mu(g) = \frac{1}{\exp(\sqrt{g/a_0}) - 1}. \quad (2)$$

This is formally identical to a Bose–Einstein occupation number with dimensionless energy $x = \sqrt{g/a_0}$.

Within Λ CDM, the RAR is explained by the interplay of dark matter halos with baryonic matter. MOND [3] postulates a modification of dynamics below a_0 but lacks a covariant formulation with acceptable properties (TeVeS [4] requires three extra fields and is constrained by GW170817).

Energy-Flow Cosmology (EFC) [5, 6] proposes that gravitational dynamics are modified by an entropy-like scalar field coupled to the spacetime substrate. This note reports a systematic attempt to derive Eq. (2) from a covariant action, documenting what succeeded, what failed, and what structural results survive independent of the specific μ -form.

1.2 Programme structure

The construction proceeded through 17 iterations (Table 1), each testing a specific hypothesis and either confirming or falsifying it. The iterative falsification structure is itself a methodological contribution: rather than presenting only successful results, we document the full elimination chain.

Table 1: Summary of the 17-iteration construction programme.

v	Test	Result	Status
1	Wave–particle duality from grid+sampling	Consistent	✓
2	Particle-based measurement feedback	Effect exists	✓
3	Planck-scale coupling strength	$\delta\rho \sim 10^{-132}$; unmeasurable	✓
4	Field-based continuous feedback	Three regimes found	✓
5	SI-unit mapping of coupling	Grid is gravitational, not EM	✓
6	RAR functional form	Formally identical to Bose–Einstein	✓
7	Which coupling reproduces RAR	Acceleration g uniquely	✓
8	Covariant action construction	Minimal action found	✓
9	GW speed (first analysis)	$c_{\text{gw}} = c$ (argument)	✓
10	Full quadratic action $\mathcal{L}^{(2)}$	No $\partial h \cdot \partial h$ from F	✓
11	Generalised Z -tensor, arbitrary $\partial_\mu \bar{\mathcal{S}}$	Speed isotropic	✓
12	Φ/Ψ slip derivation	$\eta = 1 + \mathcal{O}(\Phi\mu)$	✓
13	Self-consistent $\bar{\mathcal{S}}(r)$, $V = 0$	Tracking fails (225%)	×
14	Derive $V(\bar{\mathcal{S}})$ from tracking	$V = \frac{1}{2}m_{\bar{\mathcal{S}}}^2\bar{\mathcal{S}}^2$ restores it	✓
15	Screening mechanism	BE amplitude suppression	✓
16	Fifth force / PPN	All constraints satisfied (exp. suppressed)	✓
17	Modified Poisson from field equations	$T_{00}^{\mathcal{S}} \propto g$ (wrong direction)	×

2 The Covariant Action

2.1 Action principle

The EFC action is

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G_N} + \frac{1}{2} g^{\mu\nu} \partial_\mu \mathcal{S} \partial_\nu \mathcal{S} - V(\mathcal{S}) - \beta |\partial \mathcal{S}|_\varepsilon \mathcal{S}^2 + \mathcal{L}_m \right], \quad (3)$$

where $|\partial \mathcal{S}|_\varepsilon \equiv \sqrt{|(\partial_\mu \mathcal{S})(\partial^\mu \mathcal{S})| + \varepsilon^2}$ is a regularised gradient norm, $V(\mathcal{S}) = \frac{1}{2} m_{\mathcal{S}}^2 (\mathcal{S} - \mathcal{S}_0)^2$ is a quadratic potential stabilising the field, and $\mathcal{L}_m = \mathcal{L}_m[\psi_{\text{matter}}, g_{\mu\nu}]$ is minimally coupled matter.

2.2 Field content and parameter budget

Beyond GR, the theory introduces one scalar field $\mathcal{S}$ with:

- β : gradient–field coupling (sets acceleration scale),
- $m_{\mathcal{S}}$: field mass ($\lambda_{\mathcal{S}} = 1/m_{\mathcal{S}} \sim 50$ kpc, galactic screening length),
- ε : minimum grid tension (regulator, physically: vacuum floor).

All parameters combine into one observable scale:

$$a_0 = 1.2 \times 10^{-10} \text{ m/s}^2 \quad (\text{measured from SPARC [1]}). \quad (4)$$

2.3 Field equations

Metric variation $\delta S / \delta g^{\mu\nu} = 0$ gives the modified Einstein equations:

$$G_{\mu\nu} = 8\pi G_N (T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(\mathcal{S})}), \quad (5)$$

where $T_{\mu\nu}^{(\mathcal{S})}$ includes contributions from the kinetic term, potential, and β -coupling. $\mathcal{S}$ -field variation $\delta S / \delta \mathcal{S} = 0$ gives the scalar equation of motion. Both derive from the same action, satisfying closure condition 2 of [7].

3 Structural Result I: Gravitational Wave Speed

3.1 Statement

Theorem 1 (GW speed). *For any action of the form $S = \int \sqrt{-g} [R/16\pi G + F(g_{\mu\nu}, \mathcal{S}, \partial_\mu \mathcal{S})]$ where F contains no derivatives of the metric tensor $g_{\mu\nu}$, and matter is minimally coupled to the metric, the kinetic operator for transverse-traceless gravitational wave perturbations h_{ij}^{TT} is identical to GR. Consequently, $c_{gw} = c$ exactly within this class.*

3.2 Proof

The second variation $\delta^2 F / (\delta g^{\mu\nu} \delta g^{\alpha\beta})$ is purely algebraic in $h_{\mu\nu}$ because F depends on $g_{\mu\nu}$ only through contractions $g^{\mu\nu} \partial_\mu \mathcal{S} \partial_\nu \mathcal{S}$ and through $\sqrt{-g}$, neither of which involves $\partial_\alpha g_{\mu\nu}$.

Integration by parts cannot generate $\partial h \cdot \partial h$ terms from algebraic $h \cdot h$ terms (it can only remove derivatives, not add them). Therefore the principal symbol of the graviton wave equation comes entirely from the Einstein–Hilbert term:

$$\mathcal{L}_{\text{tensor}}^{(2)} = \frac{1}{64\pi G_N} \partial_\lambda h_{ij}^{TT} \partial^\lambda h_{ij}^{TT} + (\text{mass-like terms}). \quad (6)$$

The mass-like contributions from all four non-EH terms scale as $\mathcal{O}(\Phi^2) \lesssim 10^{-65} \text{ m}^{-2}$, giving

$$\frac{\Delta c}{c} \sim \frac{m_{\text{grav}}^2}{2\omega^2} \lesssim 10^{-54}, \quad (7)$$

which is 10^{39} times below the GW170817 bound $|\Delta c/c| < 10^{-15}$.

Corollary 1. *Anisotropic mass terms from the $\partial_\mu \bar{\mathcal{S}}$ background exist but produce direction-dependent speed corrections of the same negligible order. The proof holds for arbitrary background $\partial_\mu \bar{\mathcal{S}}$. $\square$*

3.3 Comparison

Horndeski theories with $G_4(X)R$ or $G_5(X)G_{\mu\nu}\partial^\mu\partial^\nu\phi$ modify the tensor kinetic operator and were largely killed by GW170817. The EFC action avoids this by having no $f(\mathcal{S})R$ coupling, no higher-curvature terms, and no Riemann coupling.

4 Structural Result II: Gravitational Slip

4.1 Derivation

In Newtonian gauge $ds^2 = -(1 + 2\Psi)dt^2 + (1 - 2\Phi)\delta_{ij}dx^i dx^j$, the slip parameter $\eta = \Phi/\Psi$ deviates from unity only if the $\mathcal{S}$ -field stress-energy has anisotropic stress.

The β -coupling contribution to the radial stress is:

$$T_{rr}^{(\beta)} = \beta \bar{\mathcal{S}}^2 \left[\frac{(\partial_r \bar{\mathcal{S}})^2}{|\partial_r \bar{\mathcal{S}}|} - g_{rr} |\partial_r \bar{\mathcal{S}}| \right] = \beta \bar{\mathcal{S}}^2 |\partial_r \bar{\mathcal{S}}| [1 - (1 - 2\Phi)] = 2\Phi \beta \bar{\mathcal{S}}^2 \kappa g. \quad (8)$$

The “1” cancels with “ $-g_{rr}$ ”, leaving an $\mathcal{O}(\Phi)$ suppression. The tangential stress $T_{\theta\theta}^{(\beta)}/r^2 = -\beta \bar{\mathcal{S}}^2 \kappa g$ is $\mathcal{O}(1)$. The resulting anisotropic stress $\sigma \propto \Phi \cdot \beta \bar{\mathcal{S}}^2 \kappa g$ gives:

$$\eta = 1 + \mathcal{O}(\Phi \cdot \mu). \quad (9)$$

In the weak-field galactic regime where $\Phi \sim 10^{-6}$ and $\mu \lesssim 1$, this gives $|\eta - 1| \lesssim 10^{-6}$. The result is independent of the specific form of μ and resolves the $\times 4800$ time-delay problem identified in earlier EFC versions [10], where the modification was incorrectly coupled to Φ (time-time metric component) rather than confined to Ψ (space-space). Extension to non-spherical geometries and cosmological perturbations remains to be demonstrated.

5 Structural Result III: Solar-System Compatibility

5.1 Amplitude suppression (not screening)

Matter is minimally coupled: $\mathcal{L}_m = \mathcal{L}_m[\psi, g_{\mu\nu}]$ with no direct $\mathcal{S}$ -matter vertex. The gravitational modification enters solely through $G_{\text{eff}} = G_N(1 + \mu)$. For $g \gg a_0$:

$$\mu(g) = \frac{1}{\exp(\sqrt{g/a_0}) - 1} \xrightarrow{g \gg a_0} \exp(-\sqrt{g/a_0}). \quad (10)$$

Table 2: Fifth-force ratio $\mu = F_{\text{fifth}}/F_N$ across the solar system.

Location	g [m/s ²]	$\sqrt{g/a_0}$	μ
Sun surface	274	1.5×10^6	≈ 0
Earth orbit	5.9×10^{-3}	7031	≈ 0
Saturn (Cassini)	6.5×10^{-5}	736	10^{-320}
Galaxy outskirts	1.2×10^{-10}	1	0.58

All PPN parameters are exponentially suppressed far below current experimental bounds in the solar system. The modification turns on only at $g \lesssim 100 a_0$, corresponding to galactic outskirts.

This is not screening in the chameleon/Vainshtein sense (no dynamical field-mass change or kinetic nonlinearity). It is *functional amplitude suppression*: the response function itself vanishes exponentially.

6 Structural Result IV: Stability

With the ε -regularised gradient, the kinetic coefficient for $\delta\mathcal{S}$ perturbations is

$$Z = 1 + \beta \bar{\mathcal{S}}^2 \frac{\varepsilon^2}{((\partial \bar{\mathcal{S}})^2 + \varepsilon^2)^{3/2}} > 1 \quad \Rightarrow \quad \text{no ghosts}. \quad (11)$$

The effective mass $m_{\text{eff}}^2 = m_{\mathcal{S}}^2 + 2\beta|\partial \bar{\mathcal{S}}| > 0$ prevents tachyons. The principal symbol of the $\mathcal{S}$ -field equation is the standard d'Alembertian, guaranteeing hyperbolicity. The weak energy condition $T_{00}^{(\mathcal{S})} > 0$ is satisfied for static backgrounds with $V \geq 0$ and $\beta > 0$.

7 Structural Result V: RAR as Bose–Einstein Statistics

7.1 Mathematical identification

Rewriting the McGaugh interpolation function:

$$\mu = \frac{g_{\text{obs}}}{g_{\text{bar}}} - 1 = \frac{\exp(-\sqrt{g_{\text{bar}}/a_0})}{1 - \exp(-\sqrt{g_{\text{bar}}/a_0})} = \frac{1}{\exp(\sqrt{g_{\text{bar}}/a_0}) - 1}. \quad (12)$$

This is the Bose–Einstein distribution $\langle n \rangle = 1/(\exp(E/k_B T) - 1)$ with $E/k_B T = \sqrt{g/a_0}$.

7.2 Physical interpretation

If the spacetime grid has discrete nodes with bosonic excitation modes, and the excitation energy couples to local acceleration as $E \propto \sqrt{g}$, then the gravitational enhancement is the mean occupation number at grid temperature $k_B T_{\text{grid}} \propto \sqrt{a_0}$.

7.3 Uniqueness of the acceleration coupling

Three couplings were tested against the RAR (v7):

- Volume density ρ : $m_{\text{eff}} \propto \sqrt{G\rho}$ — wrong functional form.
- Potential Φ : $m_{\text{eff}} \propto \sqrt{|\Phi|}$ — r -dependent, not universal.
- Acceleration $g = |\nabla\Phi|$: $m_{\text{eff}} \propto \sqrt{g}$ — reproduces the RAR functional form.

Only acceleration coupling gives the correct universal form. This identifies $|\nabla\Phi|$ (tidal stress on grid nodes) as the physically relevant coupling variable.

8 The Critical Gap: Classical Limit Failure

8.1 The problem

From the 00-component of the Einstein equations with the tracking solution $\partial_r \bar{\mathcal{S}} = \kappa g / c^2$, the r -dependent $\mathcal{S}$ -field energy density is:

$$T_{00}^{(\mathcal{S})}|_{\text{mod}} = \beta \bar{\mathcal{S}}_0^2 \frac{\kappa g}{c^2}. \quad (13)$$

This produces a modified Poisson equation:

$$\nabla^2 \Phi = 4\pi G_N \rho + \alpha_{\text{EFC}} \cdot g, \quad \alpha_{\text{EFC}} = \frac{4\pi G_N \beta \bar{\mathcal{S}}_0^2 \kappa}{c^4}, \quad (14)$$

where the correction term is **linear in g and increasing**.

The RAR requires the opposite: $\mu(g) \rightarrow 0$ as $g \rightarrow \infty$.

8.2 Diagnosis

The Bose–Einstein form arises from *quantum statistical mechanics* of grid excitations, not from the classical field limit. A classical local Lagrangian with finitely many derivatives generically produces polynomial corrections, not exponential suppression. This is a general limitation, not specific to the $\beta|\partial\mathcal{S}|\mathcal{S}^2$ coupling.

8.3 What survives

All structural results of Sections 3–7 are *independent* of the specific μ -form. The GW theorem depends only on the absence of ∂g in F . The slip result depends only on the $\mathcal{O}(\Phi)$ cancellation in T_{rr} . Solar-system compatibility depends only on $\mu \rightarrow 0$ at high g (any decreasing μ suffices). Stability depends only on $Z > 0$ and $m_{\text{eff}}^2 > 0$.

9 The EFT Formulation

Given the structural results (Sections 3–7) and the identified gap (Section 8), we adopt the effective field theory formulation:

$$G_{\text{eff}}(g) = G_N \left(1 + \frac{1}{\exp(\sqrt{g/a_0}) - 1} \right) \quad (15)$$

where $\mu(g)$ is postulated as an effective relation, not derived from the classical action. This EFT inherits all structural properties:

Table 3: Complete constraint summary.

Constraint	Status	Reference
$c_{\text{gw}} = c$	Shown (theorem)	Section 3
Slip $\eta \approx 1$	Shown ($\mathcal{O}(\Phi\mu)$ suppression)	Section 4
Cassini PPN	Exponentially suppressed	Section 5
LLR	Exponentially suppressed	Section 5
Pulsar timing	$\mu \approx 0$	Section 5
No ghosts	$Z > 1$	Section 6
No tachyons	$m_{\text{eff}}^2 > 0$	Section 6
Hyperbolicity	Principal symbol = $\square$	Section 6
RAR (SPARC)	Formally identical to BE	Section 7
CMB (ISW, peaks)	Not yet tested	—
$f\sigma_8(z)$	Not yet tested	—
Bullet Cluster	Not yet tested	—
Strong lensing delays	Not yet tested	—

9.1 Comparison with existing frameworks

Table 4: EFC-EFT in context.

Theory	New params	μ form	$c_{\text{gw}} = c$	Solar safe
ΛCDM	2 ($\Omega_{\text{DM}}, \Lambda$)	N/A	Yes	N/A
MOND	1 (a_0)	chosen	N/A [†]	Problematic
TeVes	3+	from vector	constrained	partial
$f(R)$	1+	from f	constrained	chameleon
Horndeski	2+	from G_i	mostly killed	Vainshtein
EFC-EFT	1 (a_0)	BE (identified)	Yes (shown)	automatic

[†]MOND is non-relativistic; covariant extensions have various c_{gw} properties.

10 Self-Consistent Field Solution

10.1 Tracking requirement

The $\mathcal{S}$ -field equation of motion (static, spherical, with $V' = 0$, $\partial_\mu n^\mu = 0$ for monotonic $\mathcal{S}$) simplifies to:

$$\mathcal{S}'' + \frac{2}{r}\mathcal{S}' - V'(\mathcal{S}) - 2\beta\mathcal{S}|\mathcal{S}'| = 0. \quad (16)$$

With $V = 0$, numerical solution in an NFW halo ($M_{\text{vir}} = 10^{12}M_\odot$, $r_s = 20$ kpc) gives tracking variation $\delta\kappa/\kappa = 225\%$ (v13) — tracking fails without a potential.

10.2 Required potential

Inserting the tracking ansatz $\mathcal{S}' = \kappa g/c^2$ into the EOM and solving for $V'(\mathcal{S})$ gives:

$$V'(\mathcal{S}) = \frac{\kappa}{c^2} [4\pi G_N \rho(r(\mathcal{S})) - 2\beta\mathcal{S} \cdot g(r(\mathcal{S}))]. \quad (17)$$

The resulting $V(\mathcal{S})$ is quadratic near $\mathcal{S}_0$:

$$V(\mathcal{S}) \approx \frac{1}{2}m_{\mathcal{S}}^2(\mathcal{S} - \mathcal{S}_0)^2, \quad m_{\mathcal{S}} \sim 10^{-28} \text{ eV}. \quad (18)$$

With this potential, tracking holds to $\sim 0.1\%$ from 1–100 kpc (v14). The quadratic potential is the EFT default — the leading renormalisable term respecting $\mathcal{S} \rightarrow -\mathcal{S}$ symmetry.

11 Methodological Contributions

11.1 Iterative falsification as research design

The 17-iteration structure (Table 1) demonstrates a constructive approach where each version tests a specific hypothesis. Failed iterations (v13: tracking without potential; v17: classical Poisson) are as valuable as successes because they precisely locate where new physics is needed.

11.2 Separation of framework from response

The most important structural insight is the clean separation between:

- **Framework** (covariant action structure, GW properties, stability) — which holds independently of μ ;
- **Response function** $\mu(g)$ — which requires microphysics not captured by the classical field equation.

This separation is analogous to hydrodynamics (Navier–Stokes) before kinetic theory, or thermodynamics before statistical mechanics: the macroscopic equations work, but the microscopic derivation of the equation of state remains open.

12 Limitations of the EFT

Several important limitations of the current framework should be stated explicitly.

First, the response function $\mu(g)$ is not derived from the classical action. The Bose–Einstein form is identified as a mathematical structure in the empirical RAR and postulated as an effective relation; the microphysical mechanism that produces this form remains unknown. Bridging this gap requires either a quantum statistical treatment of the grid degrees of freedom or a fundamentally non-local coupling, neither of which is developed here.

Second, the slip result $\eta \approx 1$ is demonstrated only in the weak-field, spherically symmetric regime. Its extension to non-spherical mass distributions, cosmological perturbation theory, and strong-field environments (neutron star mergers, black hole vicinity) has not been established.

Third, the EFT has not yet been confronted with several critical observational domains: the CMB power spectrum (ISW effect and acoustic peak positions), large-scale structure growth ($f\sigma_8$ measurements), cluster-scale lensing (Bullet Cluster-type systems), and the full cosmological background evolution $H(z)$. Any of these tests could falsify the framework.

Fourth, the theory as formulated does not address the cosmological constant problem directly. While the EFC programme motivates replacing Λ with thermodynamic evolution of the grid medium, the present EFT does not derive the background expansion history from first principles.

13 Outstanding Tests

The following observational confrontations are necessary to establish or falsify the EFT:

1. **Weak lensing consistency:** compute $\Phi + \Psi$ in a realistic galaxy model; verify $\mu_{\text{lens}} = \mu_{\text{dyn}}$ (predicted by $\eta \approx 1$).
2. **Structure growth:** predict $f\sigma_8(z)$ from the modified growth equation with $G_{\text{eff}}(z)$.
3. **CMB:** compute ISW effect and peak shifts; compare with Planck.
4. **Cluster-scale:** test whether $\mu(g)$ provides sufficient lensing mass in Bullet Cluster-type systems.
5. **Strong lensing:** verify time-delay compatibility via the $\eta \approx 1$ result.
6. **Cosmological background:** derive $H(z)$ and compare with DESI BAO/Pantheon+.

14 Conclusion

This note documents the complete construction and partial falsification of a covariant scalar–tensor model for entropy-driven gravitational modification. Five structural results survive independently of the microphysical gap:

1. $c_{\text{gw}} = c$ within the defined action class, demonstrated by theorem (Section 3).
2. Gravitational slip $\eta \approx 1$ in the weak-field galactic regime, with deviations suppressed by $\mathcal{O}(\Phi \cdot \mu)$, from a structural cancellation in T_{rr} (Section 4).
3. Solar-system constraints satisfied by exponential amplitude suppression, far below current experimental bounds (Section 5).
4. Ghost-free, tachyon-free, hyperbolic evolution (Section 6).
5. The RAR is formally identical to a Bose–Einstein occupation function with one measured scale a_0 (Section 7).

One critical negative result defines the research frontier:

6. The classical field equation produces a correction that increases with g , whereas the RAR requires a decreasing correction (Section 8).

The theory is best described as a *viable effective field theory with identified but unresolved microphysics*. The EFT formulation (Section 9) preserves all structural properties with one parameter and is ready for observational testing against lensing, structure growth, and CMB data.

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